Causal Inference under Dynamic Selection and Latent Heterogeneity

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October 27, 2026

Selection and causal panel methods

- Self-selection is a fundamental challenge in causal inference.
- Treatment and control groups are systematically different in:
  - **time-invariant** heterogeneity (fixed effects), e.g., ability;
  - **time-varying** covariates, e.g., *lagged outcomes*.
- Most causal panel methods only control for time-invariant heterogeneity.
  - e.g., difference-in-differences (DiD), synthetic control, and matrix completion,
- However, selection may also depend on time-varying covariates.

Example: Ashenfelter and Card (1985)

- **Empirical observation:** past earnings $\downarrow$ $\Rightarrow$ job training program.
- Under (i) **i.i.d.** earnings over time and (ii) **no** training effect,

$$\underbrace{\mathbb{E}(Y_{it}(0) - Y_{it-1}(0) \mid D_{it} = 1)\uparrow}_{\text{upward recovery}} > \underbrace{\mathbb{E}(Y_{it}(0) - Y_{it-1}(0) \mid D_{it} = 0)\downarrow}_{\text{downward reversion}};$$

- **Lower** past earnings + **rebounding** earnings $\Rightarrow$ positive training effect
- Similar for synthetic control and matrix completion (Arkhangelsky and Imbens, 2024).

Common practice

- In DiD: conditional parallel trend (CPT) (Abadie, 2005; Sant'Anna and Zhao, 2020):

$$\mathbb{E}\left(\mathrm{Y_{i2}}(0) - \mathrm{Y_{i1}}(0) \mid \mathrm{D_i} = 1, \mathrm{Y_{i1}}\right) = \mathbb{E}\left(\mathrm{Y_{i2}}(0) - \mathrm{Y_{i1}}(0) \mid \mathrm{D_i} = 0, \mathrm{Y_{i1}}\right),$$

- **Limitation:** when $\mathrm{Y_{it}}(0) = \alpha_\mathrm{i} + \gamma_\mathrm{t} + \epsilon_\mathrm{it}$, and $\alpha_\mathrm{i} \uparrow \Rightarrow \mathrm{D_i} = 1$ conditional on $\mathrm{Y_{i1}}$,

$$\mathrm{CPT} \Rightarrow \quad \mathbb{E}\left(\mathrm{Y_{i2}}(0) \mid \mathrm{D_i} = 1, \mathrm{Y_{i1}}\right) = \mathbb{E}\left(\mathrm{Y_{i2}}(0) \mid \mathrm{D_i} = 0, \mathrm{Y_{i1}}\right)$$

- However, treated units have larger $\alpha_\mathrm{i} \Rightarrow \mathrm{LHS} > \mathrm{RHS} \Rightarrow$ CPT fails.

Finite-sample comparisons

- DiD and doubly-robust DiD (Sant'Anna and Zhao, 2020) estimators are biased:

$$Y_{it}(0) = \alpha_i + \gamma_t + \epsilon_{it}, \quad Y_{it}(1) = Y_{it}(0) + 0.5,$$
$$D_i \sim \text{Logit}(\alpha_i - Y_{i1})$$

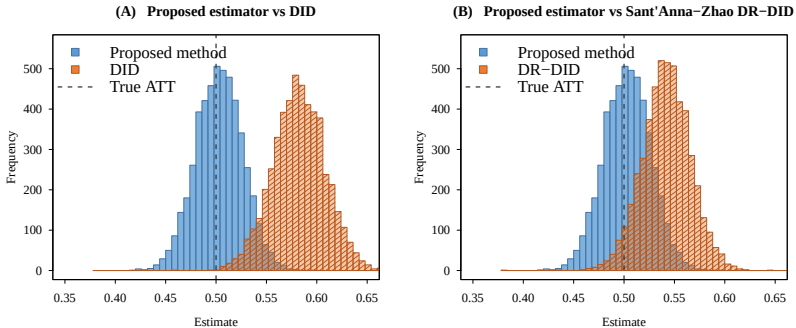


Figure: Finite-sample distributions of the proposed estimator, the DiD, and the doubly robust DiD.

This Paper

- **This paper** develops a new causal method allowing selection on (i) unobserved time-invariant heterogeneity and (ii) time-varying covariates.
- **Staggered adoption**: treatment timing may vary across units.
- Nonparametric dynamic panel:

$$Y_{it}(0) = f(X_{it}, \alpha_i, \gamma_t) + \epsilon_{it}.$$

  - **Heterogeneity**: f is an unknown function, and (α_i, γ_t) are unobserved factors.
  - **Dynamic effect**: $X_{it} := Y_{it-1}$ captures dynamic dependence (e.g., Arellano and Bond, 1991).
- Identification and doubly robust estimation of dynamic ATT with valid inference.

Overview of the method

Observation: If α_i were observed $\Rightarrow$ Match units on α and time-varying covariates.

Challenge: α_i is unobserved (and is not identified).

Solution: use pre-treatment outcomes to find units with similar α.

- Pairwise pseudo-distance: measuring similarity between units' pretreatment histories;
- *Insights*: histories are informative $\Rightarrow$ pseudo-distance as a proxy for $\|\alpha_i - \alpha_j\|$.
- Matching on pseudo-distance.

(Debias) Matching on noisy proxies $\Rightarrow$ bias in the estimator.

- Combining doubly robust estimation with *double cross-fitting* to debias.

Contribution

- Developing a new causal panel method allowing for a *general selection mechanism*.
  - Causal panel methods (De Chaisemartin and d'Haultfoeuille, 2023; Roth et al., 2023; Arkhangelsky and Imbens, 2024)
- Building on a *dynamic* panel model with rich heterogeneity.
  - Panel Econometrics literature: (e.g., Arellano and Bond, 1991; Blundell and Bond, 1998; Blundell et al., 2002; Bonhomme, 2012; Bonhomme et al., 2022).
  - Bridging the gap between the causal inference and panel econometrics literatures.
- Extending the pseudo-distance approach to dynamic models.
  - Nonlinear factor models (e.g., Feng, 2020; Athey and Imbens, 2025; Deaner et al., 2025).
- Novel *doubly cross-fitting* in causal inference.
  - DR (Arkhangelsky and Imbens, 2022; Abadie et al., 2024; Deaner et al., 2025); double cross-fitting (Newey and Robins, 2018; McClean et al., 2026).

Outline

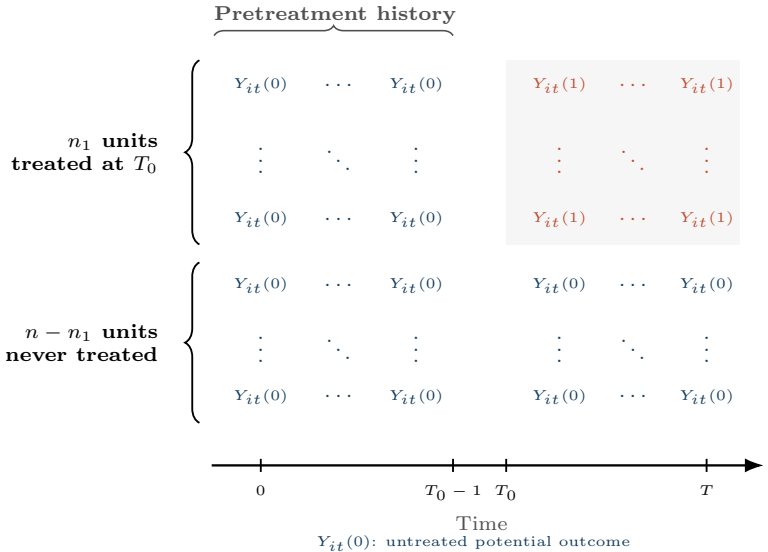
1. Setup
2. Method
3. Theory
4. Numerical Evidence
5. Empirical application

Setup

- Panel data $\{(Y_{it}, X_{it}, D_i)\}$ on n units over $t = 0, \ldots, T$ periods.
  - Y_{it} is the outcome, $D_i \in \{0, 1\}$ is the treatment.
  - $X_{it} \in \mathbb{R}^{d_X}$: predetermined time-varying covariates, e.g., $X_{it} := Y_{it-1}$.
  - $\alpha_i \in \mathbb{R}^{d_\alpha}$ is (multidimensional) unobserved heterogeneity.
- $(Y_{it}(0), Y_{it}(1))$ are potential outcomes: $Y_{it} = (1 - D_i)Y_{it}(0) + D_i Y_{it}(1)$.
- **Block assignment**: treatment occurs only at T_0 (large T_0).
- Time fixed effect γ_t: let $\Gamma_T := (\ldots, \gamma_{T-1}, \gamma_T)$ and $\mathbb{E}_T(\cdot) := \mathbb{E}(\cdot \mid \Gamma_T)$.
- **Estimand**: dynamic average treatment effect on the treated (ATT):

$$\mathrm{ATT}(t) := \mathbb{E}_T\left(Y_{it}(1) - Y_{it}(0) \mid D_i = 1\right), \quad t = T_0, \ldots, T.$$

Block assignment



Potential outcome

- **Nonparametric dynamic panel**:

$$Y_{it}(0) = f(X_{it}, \alpha_i, \gamma_t) + \epsilon_{it}, \quad \mathbb{E}_T\left(\epsilon_{it} \mid H_{it-1}, X_{it}, \alpha_i\right) = 0.$$

  Here, (i) $f(\cdot)$ is nonparametric, (iii) $H_{it} := \{(Y_{i\tau}(0), X_{i\tau})\}_{\tau \le t}$ is the history, (iii) $\{\epsilon_{it}\}$ are sequentially exogenous shocks, and (iv) $X_{it} := Y_{it-1}$ to capture dynamic effect.

- **Examples**
  - TWFE / DiD: $Y_{it}(0) = \alpha_i + \gamma_t + \epsilon_{it}$; SC, matrix completion: $Y_{it}(0) = \alpha_i' \gamma_t + \epsilon_{it}$;
  - Dynamic panel: $Y_{it} = \rho Y_{it-1} + \alpha_i + \gamma_t + \epsilon_{it}$, $Y_{it} \sim \text{Logit}(\beta Y_{it-1} + \alpha_i + \gamma_t)$.

Selection

Selection: D_i is exogenous conditional on time-varying covariates X_{iT_0} and latent factors α_i:

$$Y_{it}(0) \perp D_i \mid X_{iT_0}, \alpha_i, \Gamma_T, \quad \forall\, t = T_0, \ldots, T.$$

Common practice: If α_i were observed $\Rightarrow$ selection on observables.

(i) Define the regression outcome and propensity score as

$$m_t(x, \alpha) := \mathbb{E}_T(Y_{it}(0) \mid X_{iT_0} = x, \alpha_i = \alpha), \quad \pi(x, \alpha) := \mathbb{P}_T\left(D_i = 1 \mid X_{iT_0} = x, \alpha_i = \alpha\right).$$

(ii) Estimate $(\widehat{m}_t, \widehat{\pi})$ (nonparametrically).

(iii) Construct doubly-robust estimator:

$$\widehat{ATT(t)} := \frac{1}{n_1} \sum_{i=1}^{n} \left(D_i Y_{it} - \frac{(1 - D_i)\widehat{\pi}_i Y_{it} + (D_i - \widehat{\pi}_i)\widehat{m}_{it}}{1 - \widehat{\pi}_i} \right)$$

However, α_i is unobserved.

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Latent heterogeneity in dynamic models

- **Observation**: when $Y_{it}(0) = f(\alpha_i, \gamma_t) + \epsilon_{it}$, $\alpha_i \approx \alpha_j \Rightarrow f(\alpha_i, \gamma_t) \approx f(\alpha_j, \gamma_t) \Rightarrow$ units with similar α have *similar* histories $\{Y_{it}\}_{t=0}^{T_0-1}$ (denoised).
  - e.g., Deaner et al. (2025), Athey and Imbens (2025).
- However, dynamic dependence complicates analysis: since $Y_{it-1} \neq Y_{jt-1}$,

$$\alpha_i = \alpha_j \not\Rightarrow f(Y_{it-1}, \alpha_i, \gamma_t) = f(Y_{jt-1}, \alpha_j, \gamma_t).$$

- **Key assumption**: $\{Y_{it}(0)\}_{t \leq T}$ are weakly dependent conditional on (α_i, Γ_T). weak-dependence
- **Intuition:** dynamic effects decay over time, whereas α_i affects the entire outcome path.

A motivating example

- Consider linear panel model and iterate it backward L periods:

$$\begin{aligned} Y_{it}(0) =& \rho Y_{it-1}(0) + \alpha_i + \gamma_t + \epsilon_{it} \\ \Rightarrow Y_{it}(0) =& \rho^L Y_{it-L}(0) + \sum_{\tau=0}^{L-1} \rho^\tau (\alpha_i + \gamma_{t-\tau}) + \sum_{\tau=0}^{L-1} \rho^\tau \epsilon_{it-\tau} \end{aligned}$$

  - As $L \to \infty$, the first term $\to 0$ as long as $|\rho| < 1$.

- Even a dynamic model admits a factor model representation: $Y_{it}(0) = g(\alpha_i, \Gamma_t) + \tilde{\epsilon}_{it}$,

$$g(\alpha_i, \Gamma_t) := \sum_{\tau=0}^{\infty} \rho^\tau (\alpha_i + \gamma_{t-\tau}), \quad \tilde{\epsilon}_{it} := \sum_{\tau=0}^{\infty} \rho^\tau \epsilon_{it-\tau}.$$

A factor model representation

- Generalize the idea to nonlinear models and backward project Y_{it} onto the space of (α_i, Γ_t):

$$Y_{it}(0) = \underbrace{\mathbb{E}\left(Y_{it}(0) \mid \alpha_i, \Gamma_t\right)}_{:=g(\alpha_i,\Gamma_t)} + \underbrace{Y_{it}(0) - \mathbb{E}\left(Y_{it}(0) \mid \alpha_i, \Gamma_t\right)}_{:=\tilde{\epsilon}_{it}} = g(\alpha_i, \Gamma_t) + \tilde{\epsilon}_{it}.$$

- Dynamic models admit factor model representations with conditionally **weakly dependent** $\tilde{\epsilon}_{it}$.
- $\alpha_i \approx \alpha_j \implies g(\alpha_i, \Gamma_t) \approx g(\alpha_j, \Gamma_t) \implies$ units with similar α have *similar* histories $\{Y_{it}\}_{t=0}^{T_0-1}$.
- **Extend** the pseudo-distance to dynamic models.

Pseudo-distance

- For each i, j, the pseudo-distance $\widehat{d}_{ij}$ is obtained based on pretreatment history:

$$\widehat{d}_{ij} := \max_{1\le k_1,k_2\le n,\ k_1,k_2\neq i,j} \left| \frac{1}{T_0} \sum_{t=0}^{T_0-1} (Y_{k_1 t} - Y_{k_2 t})(Y_{it} - Y_{jt}) \right|.$$

- Time averaging smooths out transitory shocks while preserving the influence of α_i.
- Similar pretreatment histories $\Rightarrow$ pseudo distance $\widehat{d}_{ij} \approx 0$.
- Informativeness: pseudo distance $\widehat{d}_{ij} \approx 0 \Rightarrow \|\alpha_i - \alpha_j\| \approx 0$.

Estimator

- Let $m_{it} := m_t(X_{iT_0}, \alpha_i)$ and $\pi_i := \pi(X_{iT_0}, \alpha_i)$, the doubly robust estimator is:

$$\widehat{ATT(t)} := \frac{1}{n_1} \sum_{i=1}^{n} \left(D_i Y_{it} - \frac{(1 - D_i)\widehat{\pi}_i Y_{it} + (D_i - \widehat{\pi}_i)\widehat{m}_{it}}{1 - \widehat{\pi}_i} \right)$$

- $(\widehat{m}_{it}, \widehat{\pi}_i)$ are Nadaraya-Watson (NW) estimators (with bandwidths h_m, h_π)

$$\widehat{m}_{it} = \sum_{j=1}^{n} (1 - D_j) Y_{jt} \widehat{K}_{h_m,ij} \Big/ \sum_{j=1}^{n} (1 - D_j) \widehat{K}_{h_m,ij}, \quad \widehat{\pi}_i = \sum_{j=1}^{n} D_j \widehat{K}_{h_\pi,ij} \Big/ \sum_{j=1}^{n} \widehat{K}_{h_\pi,ij},$$

where $\widehat{K}_{h,ij}$ is the kernel weight (replacing $\|\alpha_i - \alpha_j\|$ with $\widehat{d}_{ij}$):

$$\widehat{K}_{h,ij} := K\left((X_{jT_0} - X_{iT_0})/h\right) K(\widehat{d}_{ij}/h).$$

Debias - Double Cross-Fitting

Debias $\widehat{\mathrm{ATT}(\mathrm{t})}$ due to (i) nonparametric estimation errors in $(\widehat{\mathrm{m}}_{\mathrm{it}}, \widehat{\pi}_{\mathrm{i}})$, and (ii) noisy $\widehat{\mathrm{d}}_{\mathrm{ij}}$.

Cross-fitting: randomly split the sample into $\{\mathcal{I}_1, \mathcal{I}_2\}$:

- $\mathcal{I}_1 \Rightarrow$ evaluating $\widehat{\mathrm{ATT}(\mathrm{t})}$, $\mathcal{I}_2 \Rightarrow$ estimating $(\widehat{\mathrm{m}}_{\mathrm{it}}, \widehat{\pi}_{\mathrm{i}})$;
- estimation errors of $(\widehat{\mathrm{m}}_{\mathrm{it}}, \widehat{\pi}_{\mathrm{i}})$ are correlated.

Double cross-fitting: randomly split the sample into $\{\mathcal{I}_1, \mathcal{I}_2, \mathcal{I}_3\}$:

- $\mathcal{I}_1 \Rightarrow$ evaluating $\widehat{\mathrm{ATT}(\mathrm{t})}$, $\mathcal{I}_2 \Rightarrow$ estimating $\widehat{\mathrm{m}}_{\mathrm{it}}$, $\mathcal{I}_3 \Rightarrow$ estimating $\widehat{\pi}_{\mathrm{i}}$;
- reduces the dependence between $\widehat{\mathrm{m}}_{\mathrm{it}}$ and $\widehat{\pi}_{\mathrm{i}}$ to debias the estimator.

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Informativeness

Assumption 1 (Informativeness sufficient conditions)

(i) ***(Population pseudo distance)*** *There exists a continuous function function* $\mathrm{d} : \mathcal{A} \times \mathcal{A} \to [0, \infty)$ *s.t. for each* $\mathrm{i}, \mathrm{j} = 1, \ldots, \mathrm{n}$,

$$\widehat{\mathrm{d}}_{\mathrm{ij}} \xrightarrow{\mathrm{p}} \mathrm{d}(\alpha_{\mathrm{i}}, \alpha_{\mathrm{j}}), \quad \mathrm{n}, \mathrm{T}_0 \to \infty.$$

(ii) ***(Informativeness)*** $\forall\ \varepsilon > 0$, *there exists a* $\delta > 0$, *such that*

$$\sup_{\alpha_1, \alpha_2 \in \mathcal{A}, \mathrm{d}(\alpha_1, \alpha_2) \leq \delta} \|\alpha_1 - \alpha_2\| \leq \varepsilon.$$

- $\widehat{\mathrm{d}}_{\mathrm{ij}},\ \mathrm{d}(\alpha_{\mathrm{i}}, \alpha_{\mathrm{j}}) \downarrow 0 \Rightarrow \|\alpha_{\mathrm{i}} - \alpha_{\mathrm{j}}\| \approx 0$, matching on $\widehat{\mathrm{d}}_{\mathrm{ij}}$ (and $\mathrm{d}(\alpha_{\mathrm{i}}, \alpha_{\mathrm{j}})$) $\Rightarrow$ matching on α.
- Example: for a TWFE model $\mathrm{Y}_{\mathrm{it}}(0) = \alpha_{\mathrm{i}} + \gamma_{\mathrm{t}} + \epsilon_{\mathrm{it}}$, $\mathrm{d}(\alpha_{\mathrm{i}}, \alpha_{\mathrm{j}}) \propto |\alpha_{\mathrm{i}} - \alpha_{\mathrm{j}}|$.

Identification

Theorem (Identification of ATT)

When $\widehat{d}_{ij}$ *is informative, then, under certain regularity conditions* assumptions, $\mathrm{ATT}(t)$ *is identified by:*

$$\mathbb{E}_T(Y_{it} \mid D_i = 1) - \mathbb{E}_T\left(\lim_{\delta \downarrow 0} \mathbb{E}_T(Y_{jt} \mid X_{jT_0} = X_{iT_0}, d(\alpha_i, \alpha_j) \leq \delta, D_j = 0) \mid D_i = 1\right).$$

- Matching: Match treated i to untreated j with similar covariates X and small pseudo distance.
- Imputation: Using matched j to impute i's counterfactual outcome $Y_{it}(0)$ by

$$\lim_{\delta \downarrow 0} \mathbb{E}_T(Y_{jt} \mid X_{jT_0} = X_{iT_0}, d(\alpha_i, \alpha_j) \leq \delta, D_j = 0)$$

- Averaging the imputed counterfactual over treated units i yields the identification result.

Theory - Imputation

Proposition (Convergence rates for nuisance)

Under regularity assumptions, suppose that for each $h \in \{h_m, h_\pi\}$, $h \to 0$, $nh^{d_\alpha + d_X}/\log n \to \infty$, *and* $\sqrt{\log(nT_0)/T_0}/h \to 0$. *Then, for each* $t \geq T_0$, assumptions

$$\max_{i=1,\ldots,n} |\widehat{m}_{it} - m_{it}| = O_P\left(h_m + \left(nh_m^{d_\alpha + d_X}\right)^{-1/2} \sqrt{\log(n)}\right).$$

In addition,

$$\max_{i=1,\ldots,n} |\widehat{\pi}_i - \pi_i| = O_P\left(h_\pi + \left(nh_\pi^{d_\alpha + d_X}\right)^{-1/2} \sqrt{\log(n)}\right).$$

- Standard bias-variance: bias $\sim h$ and std $\sim \left(nh^{d_\alpha + d_X}\right)^{-1/2} \sqrt{\log(n)}$.

Theory - Inference

Theorem (Asymptotics $\widehat{\mathrm{ATT}(t)}$)

Under regularity assumptions, for each $t = T_0, \ldots, T$, assumptions

$$\widehat{\mathrm{ATT}(t)} - \mathrm{ATT}(t) = \frac{1}{\sqrt{n}}\mathcal{N}(0, V_t) + O_P\left(h_m\left(h_\pi + (nh_\pi^{d_\alpha + d_X})^{-1/2}\sqrt{\log(n)}\right)\right) + o_P(n^{-1/2}).$$

- Double cross-fitting: a faster rate by undersmoothing h_m:
  - The variance term $(nh_m^{d_\alpha + d_X})^{-1/2}\sqrt{\log(n)}$ does not enter the remainder.
  - $\Rightarrow$ no bias-variance trade-off.

Theory - Inference

Theorem (Inference for $\widehat{\mathrm{ATT}(t)}$)

Under regularity assumptions, set $h_\pi \asymp n^{-1/(d_\alpha+d_X+2)}$ *and* $h_m \asymp n^{-1/(d_\alpha+d_X)}(\log(n))^{2/(d_\alpha+d_X)}$. *Then, for each* $t \geq T_0$ *and* $d_\alpha + d_X \leq 3$, assumptions

$$\sqrt{n}\left(\widehat{\mathrm{ATT}(t)} - \mathrm{ATT}(t)\right) \xrightarrow{d} \mathcal{N}(0, V_t).$$

- Undersmooth h_m *as aggressively as possible*. cross-validation
- Relax the dimensionality restriction from $d_\alpha + d_X \leq 1$ under standard cross-fitting to $d_\alpha + d_X \leq 3$.

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2. Method
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Simulation: DGP

- **(Linear dynamic panel)**:

$$Y_{it} = \begin{cases} \rho Y_{it-1} + \alpha_i + \gamma_t + \epsilon_{Y,it}, & t < T_0 \\ \rho Y_{it-1} + \alpha_i + \gamma_t + 0.5 \cdot D_i + \epsilon_{Y,it}, & t \geq T_0 \end{cases}, \quad D_i \sim \text{Logit}\left(\frac{\alpha_i}{2} + \frac{Y_{iT_0-1}}{2}\right)$$

  - $\rho = 0.8, \quad \alpha_i, \gamma_t \overset{\text{i.i.d.}}{\sim} U(-1/4, 1/4), \ \epsilon_{it} \overset{\text{i.i.d.}}{\sim} U(-1/2, 1/2).$

- **(Linear dynamic panel with IFE)**:

$$Y_{it} = \begin{cases} \rho Y_{it-1} + \alpha_i \gamma_t + \epsilon_{Y,it}, & t < T_0 \\ \rho Y_{it-1} + \alpha_i \gamma_t + 0.5 \cdot D_i + \epsilon_{Y,it}, & t \geq T_0 \end{cases}, \quad D_i \sim \text{Logit}\left(\frac{\alpha_i}{2} + \frac{Y_{iT_0-1}}{2}\right)$$

  - $\rho = 0.8, \quad \alpha_i \overset{\text{i.i.d.}}{\sim} U(-1, 1), \quad \gamma_t \overset{\text{i.i.d.}}{\sim} U(-2, 2), \ \epsilon_{it} \overset{\text{i.i.d.}}{\sim} U(-1/2, 1/2).$

- **(Nonlinear dynamic panel)**:

$$Y_{it} \sim \begin{cases} \text{Logit}(-2 + 0.5 \cdot Y_{it-1} + 4\alpha_i), & t < T_0 \\ \text{Logit}(-2 + 0.5 \cdot Y_{it-1} + 4\alpha_i + D_i), & t \geq T_0 \end{cases}, \quad D_i \sim \text{Logit}\left(\frac{\alpha_i}{2} + \frac{Y_{iT_0-1}}{2}\right)$$

  - $\alpha_i \overset{\text{i.i.d.}}{\sim} U(-1, 1).$

Simulation: linear dynamic model

	DiD	DR*_CF	DR_CF	DR*_DCF	DR_DCF
$N = 1000, T_0 = 20$					
Bias	2.16	0.50	0.79	0.18	0.49
SD	1.94	2.05	2.10	2.40	2.43
Coverage	79.8	93.9	92.9	94.2	93.8
$N = 1000, T_0 = 40$					
Bias	2.16	0.53	0.81	0.21	0.52
SD	1.94	2.05	2.11	2.39	2.40
Coverage	79.8	93.6	92.8	93.9	94.2
$N = 2000, T_0 = 100$					
Bias	2.16	0.36	0.55	0.14	0.37
SD	1.94	1.44	1.46	1.62	1.63
Coverage	79.8	93.8	92.5	94.0	94.2

Note: Biases and standard deviations are reported in units of 0.01.

Simulation: linear dynamic panel with IFE

	DR*_CF	DR_CF	DR*_DCF	DR_DCF
$N = 1000, T_0 = 20$				
Bias	0.60	0.82	0.24	0.46
SD	2.27	2.56	2.70	3.00
Coverage	93.5	92.1	93.6	93.1
$N = 1000, T_0 = 40$				
Bias	0.63	0.80	0.30	0.51
SD	2.29	2.49	2.70	2.87
Coverage	93.5	92.1	93.8	93.4
$N = 2000, T_0 = 100$				
Bias	0.40	0.52	0.13	0.25
SD	1.56	1.62	1.83	1.92
Coverage	93.7	92.6	94.2	93.8

Note: Biases and standard deviations are reported in units of 0.01.

Simulation: nonlinear dynamic panel

	DR*_CF	DR_CF	DR*_DCF	DR_DCF
$N = 1000, T_0 = 20$				
Bias	0.20	1.10	-0.24	-0.05
SD	2.51	2.67	3.17	3.07
Coverage	95.1	93.1	94.3	94.1
$N = 1000, T_0 = 100$				
Bias	0.21	0.63	-0.25	-0.04
SD	2.56	2.58	3.14	3.06
Coverage	95.1	94.5	94.6	94.1
$N = 2000, T_0 = 100$				
Bias	0.12	0.55	-0.06	-0.04
SD	1.76	1.77	2.06	2.05
Coverage	95.1	94.5	94.8	94.3

Note: Biases and standard deviations are reported in units of 0.01.

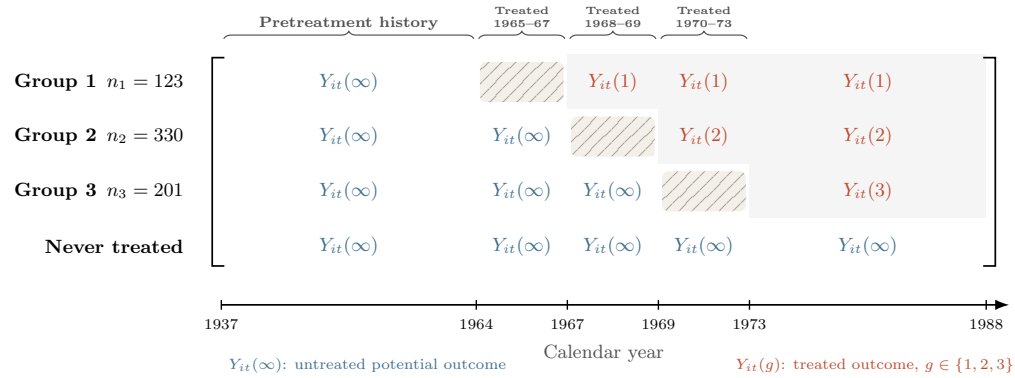
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Policy and data

- **Policy:**
  - U.S. family planning programs: federal grants subsidized contraception.
  - Treatment effects on *fertility rates* (Bailey, 2012).
- **Data:**
  - Panel: 3,017 U.S. counties from periods $1937 - 1988$.
  - Y_{it}: fertility rate, live births per $1,000$ women aged $15 - 44$.
- **Staggered adoption:**
  - Timing: first receives a federal family planning grant between $1965 - 1973$ (*early rollout*).
  - I group treated counties into three cohorts: $1965 - 67$, $1968 - 69$, and $1970 - 73$. table
- **Pretreatment history:**
  - Use fertility histories from $1937 - 1964$ to construct the pseudo-distance ($T_0 = 28$).

Staggered adoption



Figure

Selection into treatment

Local organizations application $\Longrightarrow$ Federal grants

- **Selection on fixed effects:**
  - e.g., (education and urbanization)$\uparrow$ $\Rightarrow$ application.
- **Selection on lagged fertility:**
  - **Channel 1:** recent fertility $\uparrow$ $\Rightarrow$ demand for funding $\uparrow$ $\Rightarrow$ application $\uparrow$.
  - **Channel 2:** recent fertility $\downarrow$ $\Rightarrow$ easier to claim program success $\Longrightarrow$ approval probability $\uparrow$.

Empirical evidence: selection on lagged fertility

selections

For each $g = 1, 2, 3$ and each t in group g,

$$D_{it} \sim \text{Logit}\left(\beta_g' X_{it} + \alpha_i\right);$$

$$(1) : X_{it} = Y_{it-1}, \quad (2) : X_{it} = (Y_{it-1} + Y_{it-2} + Y_{it-3})/3.$$

- α_i is *unobserved* $\Rightarrow$ proxied by *education*, *population*, etc.

	Treated 1965 − 67		Treated 1968 − 69		Treated 1970 − 73	
	(1)	(2)	(1)	(2)	(1)	(2)
$\widehat{\beta}_g$	-2.08^{***}	-1.76^{***}	0.46	0.15	0.56^{**}	0.46
	(0.56)	(0.56)	(0.34)	(0.35)	(0.27)	(0.37)
AME	−0.03	−0.02	0.02	0.01	0.01	0.01

Note: Standard errors are reported in parentheses. Coefficients, standard errors, and AME are reported in units of 0.01.

Dynamic dependence

For never-treated units and $t = 1965, \ldots, 1988$

$$Y_{it}(\infty) = \rho Y_{it-1}(\infty) + \alpha_i + \gamma_t + \epsilon_{it}.$$

- α_i is *unobserved* $\Rightarrow$ using (i) Arellano-Bond estimator (ii) dynamic panel with large T.

	Arellano–Bond			Dynamic-panel
	IV lag 2	IV lags 2–4	IV lags 2–6	
$\widehat{\rho}$	0.21^{***}	0.24^{***}	0.34^{***}	0.50^{***}
	(0.03)	(0.04)	(0.04)	(0.02)

Notes: Standard errors are reported in parentheses. The first three columns report two-step Arellano–Bond difference-GMM estimates using the indicated lag windows. The final column reports the bias-corrected dynamic-panel within estimate. $^{***}p < 0.01$.

Treatment selection may depend jointly on fixed effects and lagged outcomes.

Estimation

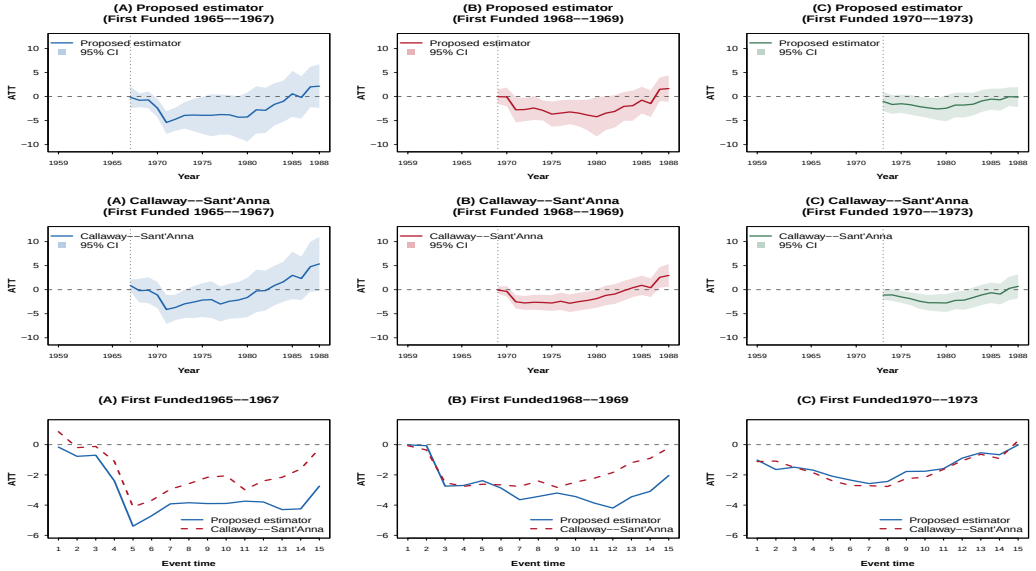
Three specifications that differ in the pretreatment history and the construction of X_{it}.

	Pretreatment history	X_{it}
Model 1	$1937 - 1964$	$(Y_{it-1} + Y_{it-2} + Y_{it-3})/3$
Model 2	$1937 - 1964$	Y_{it-1}
Model 3	$1946 - 1964$	$(Y_{it-1} + Y_{it-2} + Y_{it-3})/3$

Comparison: CS DiD estimator (Callaway and Sant'Anna, 2021).

Result - Model 1

other models



ATT by Post-Treatment Horizon

other models

- **Overall pattern:** *both* methods suggest reductions in fertility and *long-term* effects.

	ATT (1 - 5)	ATT (6 - 10)	ATT (11 - 15)
Model 1	-1.70^{**}	-3.39^{***}	-3.03^{**}
	(0.67)	(1.01)	(1.39)
CS DiD	-1.37^{**}	-2.63^{***}	-1.42
	(0.54)	(0.80)	(0.94)
Difference	-0.32	-0.76	-1.61
	(0.76)	(1.11)	(1.28)

Note: The results are weighted by the number of women aged $15-44$ in 1970.

Comparison with CS DiD

The proposed method has:

- **larger fertility reductions** for the $1965-67$ and $1968-69$ groups;
- **significant long-term effects** (11–15 years), but not under CS DiD.
- **widening gap**: the difference between the two methods' estimates grows over time.

Intuitive explanation:

- with selection on lower Y_{it-1},

$$\underbrace{\mathbb{E}(Y_{it}(0) - Y_{it-1}(0) \mid D_{it} = 1)\uparrow}_{\text{upward recovery}} > \underbrace{\mathbb{E}(Y_{it}(0) - Y_{it-1}(0) \mid D_{it} = 0)\downarrow}_{\text{downward reversion}} \Rightarrow \text{DiD} \uparrow;$$

- $\rho \in (0, 1) \Rightarrow$ gradual recovery $\Rightarrow$ larger upward DiD bias at *longer term*.

Weak Dependence

back

Assumption 2 (Weak Dependence)

$\{Y_{it}(0)\}_{t\in\mathbb{Z}}$ *are conditionally weakly dependent, such that*

$$\sum_{s=-\infty}^{\infty} |\mathrm{Cov}\left(Y_{it}(0), Y_{is}(0) \mid \alpha_i = \alpha, \Gamma_T = \Gamma\right)| < \infty$$

uniformly for any supported α*, any possible time path factor* Γ*, and any* $t \in \mathbb{Z}$*.*

- The serial dependence of the untreated outcome vanishes sufficiently fast.
- The persistence captured by the α_i and Γ_T is unrestricted.
- This assumption is mild in most applications but rules out *unit root*:

$$Y_{it}(0) = \rho Y_{it-1}(0) + \alpha_i + \gamma_t + \epsilon_{it}, \quad |\rho| = 1.$$

Identification back

Define the regression outcome and propensity score as

$$m_t(x, \alpha) := \mathbb{E}_T(Y_{it}(0) \mid X_{iT_0} = x, \alpha_i = \alpha), \quad \pi(x, \alpha) := \mathbb{P}_T\left(D_i = 1 \mid X_{iT_0} = x, \alpha_i = \alpha\right).$$

Assumption 3 (Identification)

Suppose that

(i) ***(moment)*** *the first moments of* $(Y_{it}(0), Y_{it}(1))$ *exist;*

(ii) ***(uniform continuity)*** $m_t(x, \alpha)$ *is jointly uniformly continuous in* (x, α)*;*

(iii) ***(overlap)*** $\pi(X_{iT_0}, \alpha_i) \in (0, 1)$ *a.s.*

Markovian back

Assumption 4 (Markovian)

Let $H_{iT_0-1} := \{(Y_{i\tau}, X_{i\tau})\}_{\tau \le T_0-1}$ *denote the history up to* $T_0 - 1$. *Then, for each* $t = T_0, \ldots, T$,

$$(Y_{it}(0), D_i) \perp H_{iT_0-1} \mid X_{iT_0}, \alpha_i, \Gamma_T, \quad \forall\, t = T_0, \ldots, T.$$

- Earlier history H_{iT_0-1} provides no additional information about treatment assignment and untreated outcomes.
- *Stronger* than necessary: conditional moments of $(Y_{it}(0), D_i)$ are sufficient.

Estimation

Assumption 5 (Approximation error for $\widehat{d}_{ij}$)

There exist constants $\lambda_1, \lambda_2 > 0$ such that the following inequality holds wpa1:

$$\max_{i,j \in \{1,\dots,n\}} |\widehat{d}_{ij} - d(\alpha_i, \alpha_j)| \leq \underbrace{\lambda_1 n^{-1/d_\alpha} \log(n) + \lambda_2 T_0^{-1/2} \sqrt{\log(nT_0)}}_{\delta_{n,T_0}}.$$

- The Approximation error δ_{n,T_0} consists of (i) the **matching** error and (ii) the **sampling** error.

Assumption 6 (Informativeness - estimation)

There exist constants $\underline{\eta}, \overline{\eta} > 0$ such that for any $\alpha_1, \alpha_2 \in \mathcal{A}$,

$$\underline{\eta} d(\alpha_1, \alpha_2) \leq \|\alpha_1 - \alpha_2\| \leq \overline{\eta} d(\alpha_1, \alpha_2).$$

Estimation

Assumption 7 (Regularity)

(i) ***(boundedness)*** X_{iT_0} *is uniformly bounded;*

(ii) ***(moment)*** $\forall t \geq T_0$, $\mathbb{E}_T\left(|Y_{it}(0)|^{2+\delta}\right), \mathbb{E}_T\left(|Y_{it}(1)|^{2+\delta}\right) < \infty$ *for some* $\delta > 0$*;*

(iii) ***(compactness)*** $\mathcal{A}$ *is compact. There exist constants* $\underline{c}_1, \overline{c}_1 > 0$*, such that* $\forall \alpha_0 \in \mathcal{A}, \forall r \in [0,1]$,

$$\underline{c}_1 r^{d_\alpha} \leq \mathbb{P}_T\left(\|\alpha - \alpha_0\| \leq r\right) \leq \overline{c}_1 r^{d_\alpha}.$$

There exist constants $\underline{c}_2, \overline{c}_2 > 0$ *such that for any* $(x, \alpha) \in \operatorname{supp}(X_{iT_0}, \alpha_i)$ *and* $\forall r \in [0,1]$,

$$\underline{c}_2 r^{d_X + d_\alpha} \leq \mathbb{P}_T\left(\|X_{iT_0} - x\| \leq r, \|\alpha_i - \alpha\| \leq r\right) \leq \overline{c}_2 r^{d_X + d_\alpha};$$

(iv) ***(Lipschitz)*** $\forall t \geq T_0$, $m_t(x, \alpha)$ *and* $\pi(x, \alpha)$ *are Lipschitz continuous on* $\operatorname{supp}(X_{iT_0}, \alpha_i)$*;*

(v) ***(kernel)*** $K : [-1, 1] \mapsto \mathbb{R}_+$ *is bounded. In addition,* $K(0) > 0$ *and* $K(\cdot)$ *is Lipschitz continuous.*

Pseudo-distance

Back

Lemma (Sufficient conditions for identifying $\mathrm{d}(\alpha_i, \alpha_j)$)

Suppose (i) $\{\gamma_t\}$ *is stationary and ergodic, (ii) the support of* α_i *is compact, (iii)* $Y_{it}(0)$ *has a finite second moment, i.e.,* $\mathbb{E}\left(Y_{it}^2(0) \mid \alpha_i = \alpha, \Gamma_T = \Gamma\right) < \infty$ *uniformly over supported* α *and* Γ*, (iv)* $\mathbb{E}(\sup_{\alpha \in \mathcal{A}} g^2(\alpha, \Gamma)) < \infty$*, and (v)* $g(\alpha, \Gamma)$ *is uniformly continuous in* α*, uniformly over supported* Γ*. Then, for each* $i, j = 1, \ldots, n$

$$\widehat{d}_{ij} \xrightarrow{p} d(\alpha_i, \alpha_j),$$

where

$$d(\alpha_i, \alpha_j) = \sup_{\alpha_1, \alpha_2 \in \mathcal{A}} \left| \int (g(\alpha_1, \Gamma) - g(\alpha_2, \Gamma))(g(\alpha_i, \Gamma) - g(\alpha_j, \Gamma)) \, d\mathbb{P}(\Gamma) \right|.$$

Bandwidth

back

- Choose h_m using *leave-one-out* cross-validation to minimize MSE.

- Choose h_m based on *effective sample size*:

$$n_{i,\text{eff}}(h) = \Big(\sum_{i' \in \mathcal{I}(i) \setminus \{i\}} (1 - D_{i'}) \widehat{K}_{h,ii'} \Big)^2 \Big/ \sum_{i' \in \mathcal{I}(i) \setminus \{i\}} (1 - D_{i'}) \widehat{K}^2_{h,ii'}$$

- Inference theory $\Rightarrow$ the matched neighbors for each unit $\sim (\log(n))^2$.

- Choose constants $\kappa > 0$ and $\underline{q} \in (0, 1]$ such that

$$h_m^* = \min \left\{ h > 0 : \frac{1}{n} \sum_{i=1}^{n} 1 \left(n_{i,\text{eff}}(h) > (\kappa \log(n))^2 \right) \geq \underline{q} \right\}.$$

- $(\kappa = 0.2, \underline{q} = 80\%)$ based on numerical simulations.

Staggered adoption back

Table: Treatment Timing

	Years	No. treated by year	Total
Group 1	1965 – 1967	6, 43, 74	123
Group 2	1968 – 1969	52, 278	330
Group 3	1970 – 1973	63, 75, 53, 10	201

Selection pattern

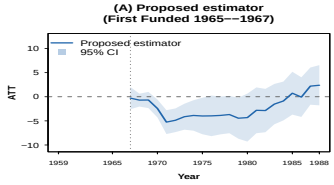
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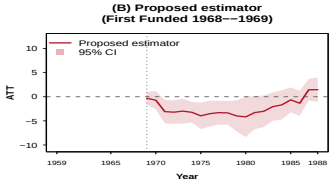
Year	Treated	Model (1)	Model (2)
Initial CAP programs			
1965	6	−5.01** (1.99)	−4.19* (2.14)
1966	43	−3.37 (2.94)	−2.41 (2.82)
1967	74	2.45 (1.56)	2.34 (1.61)
National-emphasis expansion			
1968	52	−2.45 (1.74)	−3.82* (2.14)
1969	278	0.48 (1.33)	1.27 (1.21)
Title X expansion			
1970	63	1.81 (1.34)	1.54 (1.42)
1971	75	1.99* (1.20)	3.01** (1.42)
1972	53	0.73 (2.45)	0.76 (2.81)
1973	10	−1.11 (4.56)	−1.92 (4.85)

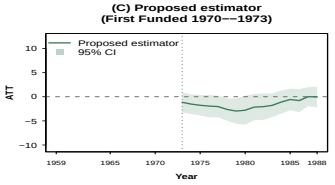
Notes: Each row reports a separate weighted Logit regression. Coefficients and standard errors, reported in parentheses, are multiplied by 100. The regressions are weighted by the number of women aged 15-44 in 1970.

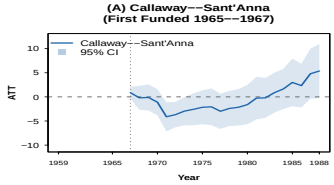
Result - Model 2

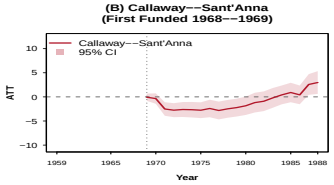
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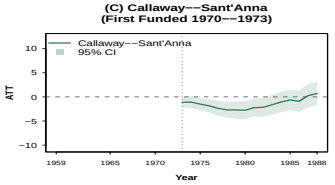


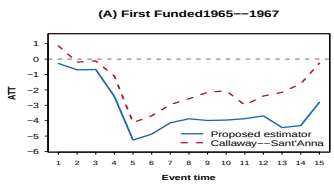


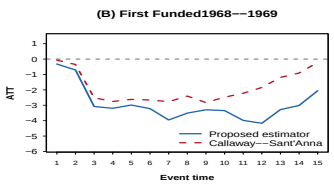


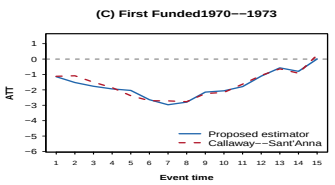






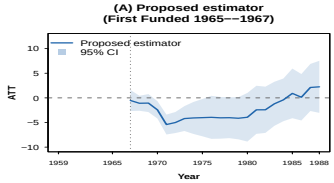


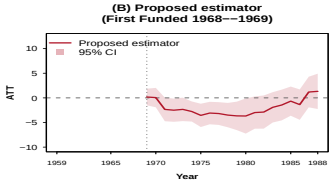


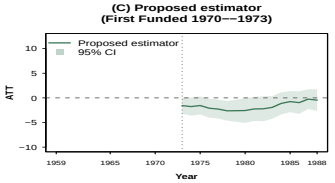


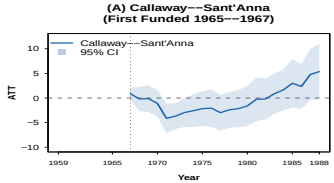
Result - Model 3

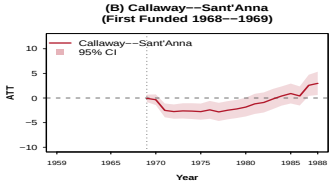
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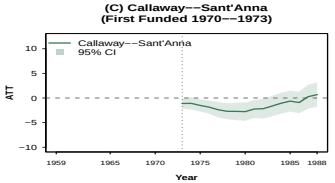


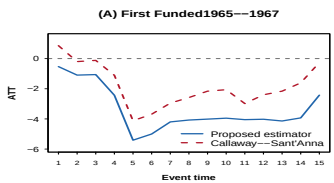


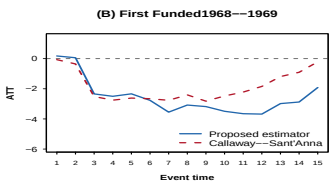


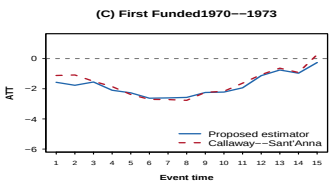












ATT by Post-Treatment Horizon

back

	ATT (1 - 5)	ATT (6 - 10)	ATT (11 - 15)
Model 1	-1.70^{**}	-3.39^{***}	-3.03^{**}
	(0.67)	(1.01)	(1.39)
Model 2	-1.74^{***}	-3.47^{***}	-2.92^{**}
	(0.59)	(0.90)	(1.36)
Model 3	-1.92^{***}	-3.56^{***}	-3.06^{**}
	(0.69)	(1.06)	(1.40)
CS DiD	-1.37^{**}	-2.63^{***}	-1.42
	(0.54)	(0.80)	(0.94)
Difference (Model 1)	-0.32	-0.76	-1.61
	(0.76)	(1.11)	(1.28)
Difference (Model 2)	-0.37	-0.89	-1.51
	(0.60)	(0.94)	(1.22)
Difference (Model 3)	-0.55	-0.93	-1.64
	(0.82)	(1.21)	(1.33)

Note: Following Bailey (2012), the results are weighted by the number of women aged $15-44$ in 1970.

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